

Tokenization & Embeddings Practice

```
pip install transformers sentencepiece
```

Birinchi, shu command orqali VSCode bilan HuggingFace ni ulavolamiz.

```
from transformers import AutoTokenizer
```

Tokenization uchun kerakli library larni va functionlarni chaqirvolamiz

```
tokenizer = AutoTokenizer.from_pretrained("gpt2")
```

Istalgan LLM modelni chaqirib AutoTokenizer orqali tokenlarga ajratamiz

```
sentence = "Ill be best LLM engineer"
```

Sentence uzgaruvchiga matn beramiz.

```
['Ill', 'Ġbe', 'Ġbest', 'ĠLL', 'ĠM', 'Ġengineer']
```

Natijada shunday qilib tokenlarga buladi.

```
tokens = tokenizer.tokenize(sentence)  
tokens
```

Shu code orqali tokenlarga ajratadi

```
number = [7457, 0]  
tokenizer.decode(number)
```

- unique ID

Shunday random raqamlarni yozib uni decode qilsa 'mistake!' shunday bulib suzlarni chiqarib beradi.

Embeddings Practice

Unique ID berdik endi ularni vectorga urzgartirish kerak.

Vector embeddings ni 2 ta layeri bor:

- **wte** (word token embedding): token ID ni **vectorga convert** qiladi
- **wpe** (word position embedding): ketma-ketligida har bir **joylashgan** tokenni **positioni** buyicha model information beradi

```
GPT2Model(  
  (wte): Embedding(50257, 768)  
  (wpe): Embedding(1024, 768)  
  (drop): Dropout(p=0.1, inplace=False)  
  (h): ModuleList(  
    (0-11): 12 x GPT2Block(  
      (ln_1): LayerNorm((768,), eps=1e-05, elementwise_affine=True)  
      (attn): GPT2Attention(  
        (c_attn): Conv1D(nf=2304, nx=768)  
        (c_proj): Conv1D(nf=768, nx=768)  
        (attn_dropout): Dropout(p=0.1, inplace=False)  
        (resid_dropout): Dropout(p=0.1, inplace=False)  
      )  
      (ln_2): LayerNorm((768,), eps=1e-05, elementwise_affine=True)  
      (mlp): GPT2MLP(  
        (c_fc): Conv1D(nf=3072, nx=768)  
        (c_proj): Conv1D(nf=768, nx=3072)  
        (act): NewGELUActivation()  
        (dropout): Dropout(p=0.1, inplace=False)  
      )  
    )  
  )  
  (ln_f): LayerNorm((768,), eps=1e-05, elementwise_affine=True)  
)
```

shu **GPT-2 modelini**
transformeri

```
print(model.wte)
```

```
Embedding(50257, 768)
```

- wte ni tekshiryapdi. 50257 bu tokenlar soni. A 768 esa model dimensional.

```
#Convert to Embeddings  
inputs = tokenizer(text, return_tensors="pt")  
  
embeddings = model.wte(inputs["input_ids"])  
  
print(embeddings.shape)  
  
torch.Size([1, 3, 768])
```

shunday qilib token ID -> Embeddings ga convert qilamiz